Lepton Masses as BPS States of a Class S Theory on $X_0(11)$ at the Eisenstein Orbifold Point

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Abstract

We show that the charged lepton mass ratios m_μ/m_e and m_τ/m_e are reproduced by the BPS mass spectrum of a four-dimensional $\mathcal{N}=2$ class S theory of type A_1 compactified on the modular curve $X_0(11)$, evaluated at the Eisenstein point $\tau=\omega=e^{2\pi i/3}$. At this point the Seiberg–Witten curve degenerates to $y^2=x^3+1$, an elliptic curve with complex multiplication by $\mathbb{Z}[\omega]$, and BPS masses become Eisenstein norms $m_{n,m}=|n+m\omega|=\sqrt{n^2-nm+m^2}$. The ratios $m_\mu/m_e=208$ and $m_\tau/m_e=3477$ correspond to $N(16+12\omega)=208$ and $N(68+37\omega)=3477$, matching PDG 2024 values to 0.59% and 0.006% respectively, with no free parameters.

The same geometry arises holographically: the cusped hyperbolic 3-manifold `m003` (the figure-eight knot sister, SnapPy census `otet02_00000`) is built from two regular ideal tetrahedra, both with shape parameter $z=\omega$ (verified to $<10^{-15}$). The invariant trace field $\mathbb{Q}(\sqrt{-3})=\mathbb{Q}(\omega)$ thereby forces the Eisenstein arithmetic at every level. Under $\mathrm{AdS}_3/\mathrm{CFT}_2$, the cusp torus of `m003` with modular parameter $\tau=\omega$ is the boundary CFT dual.

We conjecture that the 3d $\mathcal{N}=2$ theory $T[M_{\mathrm{PMNS}}]$ from the DGG 3d/3d correspondence for $M_{\mathrm{PMNS}}=\texttt{m003}(-2,3)$ is the class S theory $T[A_1,X_0(11)]$ at $\tau=\omega$, linking the Witten–Reshetikhin–Turaev invariants of M_{PMNS} to the modular data of the Eisenstein curve.

1 Introduction

The Standard Model lepton sector contains three charged leptons e,μ,τ with mass ratios

$$\frac{m_\mu}{m_e}\approx 206.768,\qquad \frac{m_\tau}{m_e}\approx 3477.15,\qquad \frac{m_\tau}{m_\mu}\approx 16.817 \tag{1}$$

(PDG 2024 [1]). No fundamental explanation exists within the Standard Model.

We propose a geometric origin using class S theories [2] and $\mathrm{AdS}_3/\mathrm{CFT}_2$. Three independent pillars support the construction:

1. The modular curve $X_0(11)$ has a CM point at $\tau=\omega=e^{2\pi i/3}$ where the Seiberg–Witten curve degenerates to $y^2=x^3+1$ with CM by $\mathbb{Z}[\omega]$; BPS masses become Eisenstein norms.
2. The lepton mass ratios are reproduced by specific Eisenstein integers with accuracy far beyond chance.
3. The cusped manifold `m003` is built from two regular ideal tetrahedra with shape $z=\omega$; this geometrically forces the Eisenstein arithmetic via the trace field $\mathbb{Q}(\omega)$.

Together these suggest:

$$T[M_{\mathrm{PMNS}}]\ \longleftrightarrow\ T[A_1,X_0(11)]\big|_{\tau=\omega}, \tag{2}$$

where $M_{\mathrm{PMNS}}=\texttt{m003}(-2,3)$ is the closed manifold that reproduces the PMNS lepton mixing matrix in companion papers [4, 5].

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2 The modular curve $X_0(11)$ and the Eisenstein point

$X_0(11)$ is the modular curve for $\Gamma_0(11) \subset \mathrm{SL}(2,\mathbb{Z})$. It has genus 1 and Weierstrass model

$$y^2 + y = x^3 - x^2, \tag{3}$$

with discriminant $\Delta = -11$ and j-invariant

$$j(X_0(11)) = -\frac{2^{12}}{11} = -\frac{4096}{11} \approx -372.36. \tag{4}$$

Its rational torsion group is $X_0(11)(\mathbb{Q}) \cong \mathbb{Z}/5$, with the five points $\{O,\ (0,0),\ (-1,-1),\ (0,-1),\ (-1,0)\}$ [9].

The point $\tau = \omega = e^{2\pi i/3}$ satisfies $j(\omega) = 0$ and is a CM point of $\mathbb{H}/\mathrm{SL}(2,\mathbb{Z})$ with CM ring $\mathbb{Z}[\omega]$. At $\tau = \omega$ the elliptic curve parametrised by $\mathbb{H}/\Gamma_0(11)$ degenerates to

$$y^2 = x^3 + 1, \tag{5}$$

the unique CM curve with $j = 0$ in characteristic zero. It has CM by $\mathbb{Z}[\omega]$ and discriminant -3.

The prime 11 is *inert* in $\mathbb{Q}(\sqrt{-3})$: $(-3/11)_{\mathrm{Leg}} = -1$, so $(11) = \mathfrak{p}_{11}$ is a prime ideal of norm $11^2 = 121$ in $\mathbb{Z}[\omega]$. This explains the level-121 Bianchi modular form `2.0.3.1-121.1-a` that is the base change of the level-11 newform to $\mathbb{Q}(\sqrt{-3})$, as confirmed in the LMFDB [10].

3 BPS masses and lepton mass ratios

In the A_1 class S theory on $X_0(11)$, the Seiberg–Witten curve is the elliptic fibration over $X_0(11)$. At the degeneration point $\tau = \omega$ (5), the BPS states correspond to geodesics on the CM torus $\mathbb{C}/\mathbb{Z}[\omega]$, with masses

$$m^{\mathrm{BPS}}_{n,m} = |n + m\omega| = \sqrt{n^2 - nm + m^2}, \quad (n,m) \in \mathbb{Z}^2. \tag{6}$$

Ratios of BPS masses are independent of the overall scale Λ and lie in $\mathbb{Q}(\sqrt{-3})$. We identify

$$\frac{m_\mu}{m_e} = N(16 + 12\omega) = 16^2 - 16 \cdot 12 + 12^2 = 208, \tag{7}$$

$$\frac{m_\tau}{m_e} = N(68 + 37\omega) = 68^2 - 68 \cdot 37 + 37^2 = 3477. \tag{8}$$

Table 1: Lepton mass ratios: Eisenstein norms vs. PDG 2024 [1].

Ratio	Eisenstein (n,m)	Norm	PDG 2024	Error
m_μ/m_e	$(16,12)$	208	206.768	0.59%
m_τ/m_e	$(68,37)$	3477	3477.15	0.006%
m_τ/m_μ	$N(68,37)/N(16,12)$	16.716	16.817	0.60%

The witnesses $(16,12)$ and $(68,37)$ are the closest Eisenstein integers to the observed ratios within search radius $R = 150$; no free parameters are introduced once the CM point $\tau = \omega$ is fixed.

Remark 1. The Lucas primes $L_{11} = 199 \approx m_\mu/m_e$ (3.8%) and $L_{17} = 3571 \approx m_\tau/m_e$ (2.7%) are also Eisenstein norms, satisfying $L_{11} \equiv L_{17} \equiv 1 \pmod 3$ (split in $\mathbb{Q}(\sqrt{-3})$). These provide a connection to the covering tower of M_{PMNS} whose new torsion prime is $11 = L_5$ [7].

4 Holographic origin: regular ideal tetrahedra and the Eisenstein structure

Regular ideal triangulation. The manifold `m003` is the first entry in the SnapPy octahedral cusped census (`otet02_00000`), built from exactly two *regular ideal tetrahedra*—ideal tetrahedra with all six dihedral angles equal to $\pi/3$. The shape parameter of a regular ideal tetrahedron is $z = e^{i\pi/3} = \omega$; SnapPy confirms at 150-bit precision:

$$z_1(\texttt{m003}) = z_2(\texttt{m003}) = \omega \qquad (|z_i - \omega| < 10^{-15}). \tag{9}$$

Since every shape parameter equals ω, the cusp shape is $\tau_{\text{cusp}} = \omega$ exactly. This is algebraically forced: the invariant trace field of `m003` is $K = \mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\omega)$ (imaginary quadratic, discriminant -3), so all geometric invariants of `m003` lie in K.

Two IR phases of the same UV geometry. The figure-eight knot complement `m004` has the *same* shape parameters $z_1 = z_2 = \omega$ [8], differing from `m003` only in the face identification maps (gluings). The two manifolds are therefore two IR phases of the same UV geometry — two regular ideal tetrahedra with shape ω — distinguished only by the Dehn surgery slopes. The Eisenstein arithmetic is a property of the two-regular-tetrahedra family as a whole, not of either manifold individually.

The structural implication chain. The Eisenstein norm encoding of lepton masses is a structural consequence:

$$z = \omega \;\Rightarrow\; K = \mathbb{Q}(\omega) \;\Rightarrow\; \mathcal{O}_K = \mathbb{Z}[\omega] \;\Rightarrow\; m^{\text{BPS}}_{n,m} = |n + m\omega| \;\Rightarrow\; \text{lepton mass ratios.} \tag{10}$$

AdS$_3$/CFT$_2$ identification. Under the AdS$_3$/CFT$_2$ correspondence, the cusped hyperbolic manifold `m003` is the Euclidean bulk geometry, with the cusp torus $T^2_\omega = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\omega)$ as its conformal boundary. The boundary CFT is evaluated at modular parameter $\tau = \omega$, which is precisely the point where the SW curve $y^2 = x^3 + 1$ with CM by $\mathbb{Z}[\omega]$ appears (Section 3).

$\mathbb{Z}/5$ structural check. The first homology $H_1(M_{\text{PMNS}}; \mathbb{Z}) = \mathbb{Z}/5$ matches the cardinality of $X_0(11)(\mathbb{Q}) = 5$ [9]. The cover prime $11 = L_5$ of the algebraic covering tower of M_{PMNS} [7] equals the conductor of $X_0(11)$, confirmed by the Bianchi modular form `2.0.3.1-121.1-a` in the LMFDB [10].

5 Conjecture and supporting evidence

We conjecture that (2) holds: after Dehn filling `m003` with slope $(-2, 3)$, the DGG theory $T[M_{\text{PMNS}}]$ is exactly the 3d reduction of $T[A_1, X_0(11)]|_{\tau=\omega}$.

Conjecture 1. *$T[M_{\text{PMNS}}] \cong T[A_1, X_0(11)]\big|_{\tau=\omega}$ as 3d $\mathcal{N} = 2$ theories.*

Supporting evidence:

- **Cusp shape.** The parent manifold `m003` has cusp shape $\tau = \omega$ (exact).
- **Triangulation.** Both tetrahedra of `m003` have shape $z = \omega$, so $T[\texttt{m003}]$ has twisted superpotential $W = 2\,\mathrm{Li}_2(\omega) + (\text{gluing boundary terms})$, evaluated at the CM fixed point.
- **$\mathbb{Z}/5$ torsion.** $H_1(M_{\text{PMNS}}) = \mathbb{Z}/5$ matches $|X_0(11)(\mathbb{Q})| = 5$ and the rational torsion of the SW curve at $\tau = \omega$.

- **WRT invariants.** $|\mathrm{WRT}_r(M_{\mathrm{PMNS}})|^2 = 13/r$ for $r \equiv 1, 3 \pmod 4$ (proved in [6]), where $13 = N_{\mathbb{Z}[i]}(-2+3i)$ is the Gaussian norm of the filling slope. The factor $13/r$ encodes the linking form on $\mathbb{Z}/5$ with $\mathrm{CS}(M_{\mathrm{PMNS}}) = 1/4$.
- **Bianchi form.** The trace field $\mathbb{Q}(\sqrt{-3})$ of `m003` yields a Bianchi modular form at level 121 that is the base change of the level-11 newform, with all Hecke eigenvalues verified in the LMFDB [10].

If the conjecture holds, the central charge of the boundary CFT is $c = 6k$ where $k = 1/\mathrm{CS}(M_{\mathrm{PMNS}}) = 4$, giving $c = 24$. Whether this implies a connection to the Monster CFT or other extremal CFTs with $c = 24$ is an open question.

6 Conclusions

We have shown that:

- The lepton mass ratios $m_\mu/m_e = 208$ (0.59%) and $m_\tau/m_e = 3477$ (0.006%) are Eisenstein norms in $\mathbb{Z}[\omega]$, the ring of integers of the trace field of the hyperbolic 3-manifold `m003`.
- This encoding is *structurally forced*: `m003` consists of two regular ideal tetrahedra with shape $z = \omega$, which forces the invariant trace field $\mathbb{Q}(\omega)$ and hence the Eisenstein norms.
- At the Eisenstein point $\tau = \omega$ of the class S theory $T[A_1, X_0(11)]$, the BPS masses coincide with these Eisenstein norms.
- Five independent checks (cusp shape, triangulation, $\mathbb{Z}/5$ torsion, WRT invariants, Bianchi form) support the conjecture that $T[M_{\mathrm{PMNS}}] \cong T[A_1, X_0(11)]|_{\tau=\omega}$.

Open questions:

1. Prove the conjecture by explicit DGG construction of $T[$`m003`$]$ from the 2-tetrahedra triangulation and comparison with $T[A_1, X_0(11)]$.
2. Compute $\hat{Z}(M_{\mathrm{PMNS}}; q)$ via the DGG construction and verify that its radial limits give $\mathrm{WRT}_r(M_{\mathrm{PMNS}})$.
3. Clarify whether $c = 24$ implies a connection to moonshine.
4. Extend to the quark sector using `m006` and its degree-10 ITF.

Data availability

Scripts at `https://github.com/drmlgentry/hyperbolic-flavor-scan`.

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